

Simple Gibbs Sampler Notes

M08: Lecture 2

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Simple Gibbs sampler

Chain binomial model

- Let's go back to the chain binomial example from the motivating examples section
- The multinomial complete data likelihood for q (the probability of failure) is:

$$f(n_1, n_{11}, N_3, n_{111}|q) = \binom{334}{n_1, n_{11}, n_{111}, N_3 - n_{111}} (q^2)^{n_1} (2q^2p)^{n_{11}} (2qp^2)^{n_{111}} (p^2)^{N_3 - n_{111}}$$

where: $N_3 = n_{111} + n_{12}$
 $= 275$

- We do not observe either n_{111} or n_{12} , just their total N_3
- The Gibbs sampler iteratively samples the model unknowns from a sequence of full conditional distributions
- To sample one draw from each full conditional distribution at each iteration, it assumes that all other model quantities are known at that iteration
- The Gibbs sampler converges to the posterior distribution of the model unknowns
- Here, n_{111} should be augmented, and the posterior distribution of q estimated.
- The joint distribution of the observations and the model unknowns is:

$$f(n_1, n_{11}, N_3, n_{111}, q) = \underbrace{f(n_1, n_{11}, N_3, n_{111}|q)}_{\text{complete data likelihood}} \times \underbrace{f(q)}_{\text{prior}}$$

- We want to make inference about the joint distribution of the model unknowns
– $f(n_{111}, q|n_1, n_{11}, N_3)$

- Possible by using the Gibbs sampler to sample from the full conditionals

- * $f(q|n_1, n_{11}, N_3, n_{111})$
- * $f(n_{111}|n_1, n_{11}, N_3, q)$

Process

- Start with initial values for unknowns
- For $t = 1 \rightarrow M$:
 - Sample $q^{(t+1)} \sim f(q|n_1, n_{11}, N_3, n_{111}^{(t)})$
 - Sample $n_{111}^{(t+1)} \sim f(n_{111}|n_1, n_{11}, N_3, q^{(t+1)})$

Full conditionals

- Getting the full conditionals is the tricky bit, but we can calculate if we assume the other unknown value are actually known

n_{111}

- Assume q is known
- The conditional probability for the chain $1 \rightarrow 1 \rightarrow 1$

$$\begin{aligned}
 \Pr(1 \rightarrow 1 \rightarrow 1 | N = 3, q) &= \frac{\Pr(N = 3, 1 \rightarrow 1 \rightarrow 1 | q)}{\Pr(N = 3 | q)} \\
 &= \frac{\Pr(N = 3 | 1 \rightarrow 1 \rightarrow 1, q) \Pr(1 \rightarrow 1 \rightarrow 1 | q)}{\Pr(N = 3 | 1 \rightarrow 1 \rightarrow 1, q) \Pr(1 \rightarrow 1 \rightarrow 1 | q) + \Pr(N = 3 | 1 \rightarrow 2, q) \Pr(1 \rightarrow 2 | q)} \\
 &= \frac{2p^2q}{2p^2q + p^2} \\
 &= \frac{2q}{2q + 1}, (0 \leq q \leq 1)
 \end{aligned}$$

$\therefore$

full conditional distribution of n_{111} :

$$n_{111} | (n_1, n_{11}, N_3, q) \sim \text{Binomial} \left(275, \frac{2q}{2q + 1} \right)$$

q

- Assume n_{111} is known
- Assume a Beta prior distribution for q
 - $q \sim \text{Beta}(\alpha, \beta)$
 - $f(q) \equiv f(q|\alpha, \beta) \propto q^{\alpha-1}(1-q)^{\beta-1}$
- The full conditional distribution of q :

$$\begin{aligned}
 f(q|n_1, n_{11}, N_3, n_{111}, \alpha, \beta) &\propto f(n_1, n_{11}, N_3, n_{111} | q, \alpha, \beta) f(q | \alpha, \beta) \\
 &\propto \underbrace{q^{2n_1+2n_{11}+n_{111}} p^{n_{11}+2N_3}}_{\text{complete data likelihood}} \times \underbrace{q^{\alpha-1}(1-q)^{\beta-1}}_{\text{prior}}
 \end{aligned}$$

$$q | \text{complete data}, \alpha, \beta \sim \text{Beta}(2n_1 + 2n_{11} + n_{111} + \alpha, n_{11} + 2N_3 + \beta)$$

- If we set $\alpha = \beta = 1$, we have a uniform prior on q
- A natural point estimate would be the mean of the Beta distribution
 - $\frac{2n_1+2n_{11}+n_{111}+\alpha}{2n_1+3n_{11}+3n_{111}+2n_{12}+\alpha+\beta}$
 - Proportion of “escapes” from all exposures

Sampling with the Gibbs sampler

Now we have the full conditionals, we can update our process from earlier:

- Start with initial values for unknowns
- For $t = 1 \rightarrow M$:
 - Sample $q^{(t+1)} \sim \text{Beta}(2n_1 + 2n_{11} + n_{111}^{(t)} + \alpha, n_{11} + 2N_3 + \beta)$
 - Sample $n_{111}^{(t+1)} \sim \text{Binomial}\left(275, \frac{2q^{(t+1)}}{2q^{(t+1)}+1}\right)$
- Get summaries of the marginal posterior distributions